

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \frac{a-b \cos 2x}{x^2} & ; \quad x < 0 \\ x^2 + cx + 2 & ; \quad 0 \leq x \leq 1 \\ 2x + 1 & ; \quad x > 1 \end{cases}$$

If f is continuous everywhere in $\mathbb{R}$ and m is the number of points where f is NOT differential then $m + a + b + c$ equals :

(1) 1

(2) 4

(3) 3

(4) 2

Q2 - 2024 (01 Feb Shift 2)

Let $f(x) = |2x^2 + 5|x| - 3|$, $x \in \mathbb{R}$. If m and n denote the number of points where f is not continuous and not differentiable respectively, then $m + n$ is equal to :

(1) 5

(2) 2

(3) 0

(4) 3

Q3 - 2024 (27 Jan Shift 1)

Consider the function.

$$f(x) = \begin{cases} \frac{a(7x-12-x^2)}{b|x^2-7x+12|} & , x < 3 \\ 2 \frac{\sin(x-3)}{x-[x]} & , x > 3 \\ b & , x = 3 \end{cases}$$

Where $[x]$ denotes the greatest integer less than or equal to x . If S denotes the set of all ordered pairs (a, b) such that $f(x)$ is continuous at $x = 3$, then the number of elements in S is :

(1) 2

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(2) Infinitely many

(3) 4

(4) 1

Q4 - 2024 (27 Jan Shift 2)

Consider the function $f : (0, 2) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{2} + \frac{2}{x}$ and the function $g(x)$ defined by

$$g(x) = \begin{cases} \min\{f(t)\}, & 0 < t \leq x \text{ and } 0 < x \leq 1 \\ \frac{3}{2} + x, & 1 < x < 2 \end{cases} \text{ . Then}$$

(1) g is continuous but not differentiable at $x = 1$ (2) g is not continuous for all $x \in (0, 2)$ (3) g is neither continuous nor differentiable at $x = 1$ (4) g is continuous and differentiable for all $x \in (0, 2)$

Q5 - 2024 (29 Jan Shift 2)

Let $f(x) = \sqrt{\lim_{r \rightarrow x} \left\{ \frac{2r^2[(f(r))^2 - f(x)f(r)]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right\}}$ be differentiable in $(-\infty, 0) \cup (0, \infty)$ and $f(1) = 1$.

Then the value of ea , such that $f(a) = 0$, is equal to _____.

Q6 - 2024 (30 Jan Shift 1)

If the function $f(x) = \begin{cases} \frac{1}{|x|}, & |x| \geq 2 \\ ax^2 + 2b, & |x| < 2 \end{cases}$ is differentiable on $\mathbb{R}$, then $48(a + b)$ is equal to _____

Q7 - 2024 (30 Jan Shift 2)

Let a and b be real constants such that the function f defined by $f(x) = \begin{cases} x^2 + 3x + a, & x \leq 1 \\ bx + 2, & x > 1 \end{cases}$ be

differentiable on $\mathbb{R}$. Then, the value of $\int_{-2}^2 f(x) dx$ equals

(1) $\frac{15}{6}$ (2) $\frac{19}{6}$

(3) 21

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(4) 17

Q8 - 2024 (30 Jan Shift 2)

Let $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ be a function satisfying $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ for all $x, y, f(y) \neq 0$. If $f'(1) = 2024$, then

(1) $xf'(x) - 2024f(x) = 0$

(2) $xf f'(x) + 2024f(x) = 0$

(3) $xf'(x) + f(x) = 2024$

(4) $xf'(x) - 2023f(x) = 0$

Q9 - 2024 (31 Jan Shift 1)

Let $g(x)$ be a linear function and $f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}, & x > 0 \end{cases}$, is continuous at $x = 0$. If $f'(1) = f(-1)$,

then the value of $g(3)$ is

(1) $\frac{1}{3} \log_e \left(\frac{4}{9e^{1/3}} \right)$

(2) $\frac{1}{3} \log_e \left(\frac{4}{9} \right) + 1$

(3) $\log_e \left(\frac{4}{9} \right) - 1$

(4) $\log_6 \left(\frac{4}{9e^{1/3}} \right)$

Q10 - 2024 (31 Jan Shift 2)

Consider the function $f : (0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = e^{-|\log_6 x|}$. If m and n be respectively the number of points at which f is not continuous and f is not differentiable, then $m + n$ is

(1) 0

(2) 3

(3) 1

(4) 2

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Questions with Answer Keys

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Answer Key

Q1 (4)	Q2 (4)	Q3 (4)	Q4 (1)		
Q5 (2)	Q6 (15)	Q7 (4)	Q8 (1)		
Q9 (4)	Q10 (3)				

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Solutions

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Q1

At $x = 1$, $f(x)$ is continuous therefore,

$$f(1^-) = f(1) = f(1^+)$$

$$f(1) = 3 + c$$

$$f(1^+) = \lim_{h \rightarrow 0} 2(1+h) + 1$$

$$f(1^+) = \lim_{h \rightarrow 0} 3 + 2h = 3$$

from (1) & (2)

$$c = 0$$

at $x = 0$, $f(x)$ is continuous therefore,

$$f(0^-) = f(0) = f(0^+)$$

$$f(0) = f(0^+) = 2$$

$f(0^-)$ has to be equal to 2

$$\lim_{h \rightarrow 0} \frac{a - b \cos(2h)}{h^2}$$

$$\lim_{h \rightarrow 0} \frac{a - b \left\{ 1 - \frac{4h^2}{2!} + \frac{16h^4}{4!} + \dots \right\}}{h^2}$$

$$\lim_{h \rightarrow 0} \frac{a - b + b \left\{ 2h^2 - \frac{2}{3}h^4 \dots \right\}}{h^2}$$

for limit to exist $a - b = 0$ and limit is $2b$

from (3), (4) & (5)

$$a = b = 1$$

checking differentiability at $x = 0$

$$\text{LHD : } \lim_{h \rightarrow 0} \frac{\frac{1 - \cos 2h}{h^2} - 2}{-h}$$

$$\lim_{h \rightarrow 0} \frac{1 - \left(1 - \frac{4h^2}{2!} + \frac{16h^4}{4!} \dots \right) - 2h^2}{-h^3} = 0$$

$$\text{RHD : } \lim_{h \rightarrow 0} \frac{(0+h)^2 + 2 - 2}{h} = 0$$

Function is differentiable at every point in its domain

$$\therefore m = 0$$

$$m + a + b + c = 0 + 1 + 1 + 0 = 2$$

Q2

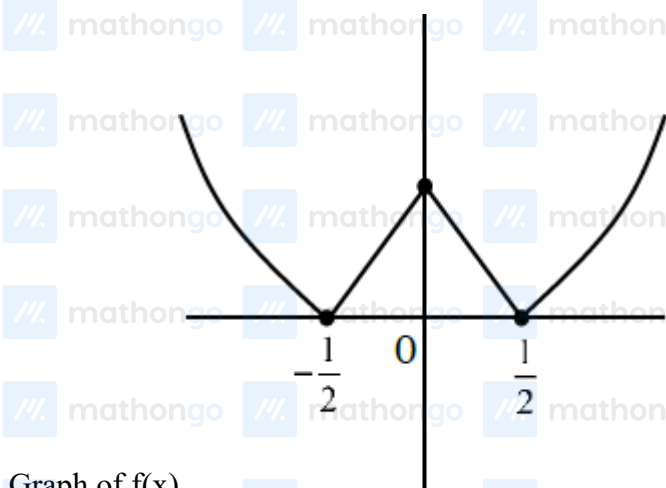
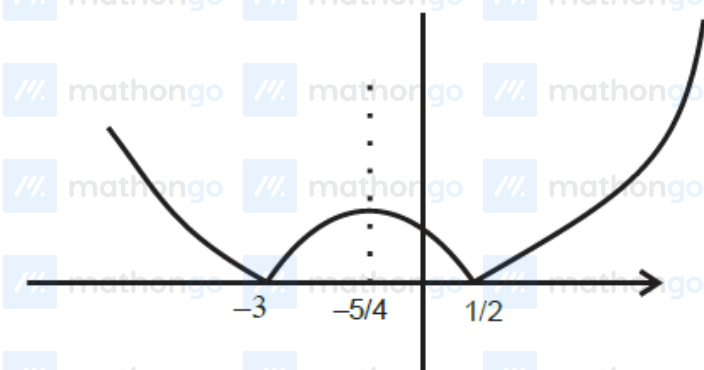
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Solutions

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$$f(x) = |2x^2 + 5|x| - 3|$$

Graph of $y = 12x^2 + 5x - 3$ Graph of $f(x)$

Number of points of discontinuity = 0 = m

Number of points of non-differentiability = 3 = n

Q3

$$f(3^-) = \frac{a}{b} \frac{(7x-12-x^2)}{|x^2-7x+12|} \quad (\text{for } f(x) \text{ to be cont.})$$

$$\Rightarrow f(3^-) = \frac{-a}{b} \frac{(x-3)(x-4)}{(x-3)(x-4)}; x < 3 \Rightarrow \frac{-a}{b}$$

$$\text{Hence } f(3^-) = \frac{-a}{b}$$

$$\text{Then } f(3^+) = 2 \lim_{x \rightarrow 3^+} \left(\frac{\sin(x-3)}{x-3} \right) = 2 \text{ and}$$

$$f(3) = b$$

$$\text{Hence } f(3) = f(3^+) = f(3^-)$$

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Solutions

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$$\Rightarrow b = 2 = -\frac{a}{b}$$

$$b = 2, a = -4$$

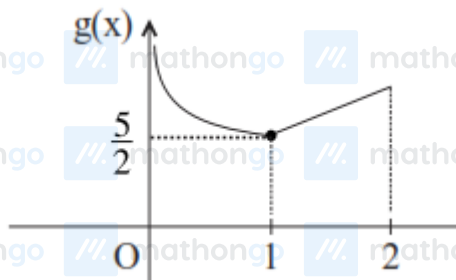
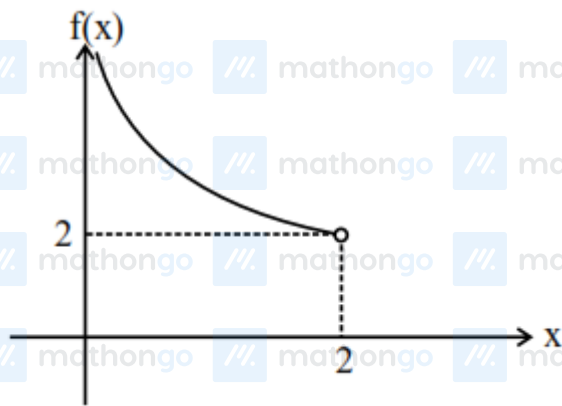
Hence only 1 ordered pair $(-4, 2)$.

Q4

$$f : (0, 2) \rightarrow \mathbb{R}; f(x) = \frac{x}{2} + \frac{2}{x}$$

$$f'(x) = \frac{1}{2} - \frac{2}{x^2}$$

$\therefore f(x)$ is decreasing in domain.



$$g(x) = \begin{cases} \frac{x}{2} + \frac{2}{x} & 0 < x \leq 1 \\ \frac{3}{2} + x & 1 < x < 2 \end{cases}$$

Q5

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Solutions

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$$f(1)=1, f(a)=0$$

$$f^2(x) = \lim_{r \rightarrow x} \left(\frac{2r^2 (f^2(r) - f(x)f(r))}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right)$$

$$= \lim_{r \rightarrow x} \left(\frac{2r^2 f(r)}{r+x} \frac{(f(r) - f(x))}{r-x} - r^3 e^{\frac{f(r)}{r}} \right)$$

$$f^2(x) = \frac{2x^2 f(x)}{2x} f'(x) - x^3 e^{\frac{f(x)}{x}}$$

$$y^2 = xy \frac{dy}{dx} - x^3 e^{\frac{y}{x}}$$

$$\frac{y}{x} = \frac{dy}{dx} - \frac{x^2}{y} e^{\frac{y}{x}}$$

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v = v + x \frac{dv}{dx} - \frac{x}{v} e^v$$

$$\frac{dv}{dx} = \frac{e^v}{v} \Rightarrow e^{-v} v dv = dx$$

Integrating both side

$$e^v (x+c) + 1 + v = 0$$

$$f(1) = 1 \Rightarrow x = 1, y = 1$$

$$\Rightarrow c = -1 - \frac{2}{e}$$

$$e^v \left(-1 - \frac{2}{e} + x \right) + 1 + v = 0$$

$$e^{\frac{y}{x}} \left(-1 - \frac{2}{e} + x \right) + 1 + \frac{y}{x} = 0$$

$$x = a, y = 0 \Rightarrow a = \frac{2}{e}$$

$$ae = 2$$

Q6

$$f(x) \begin{cases} \frac{1}{x}; x \geq 2 \\ ax^2 + 2b; -2 < x < 2 \\ -\frac{1}{x}; x \leq -2 \end{cases}$$

$$\text{Continuous at } x = 2 \Rightarrow \frac{1}{2} = \frac{a}{4} + 2b$$

$$\text{Continuous at } x = -2 \Rightarrow \frac{1}{2} = \frac{a}{4} + 2b$$

Since, it is differentiable at $x = 2$

$$-\frac{1}{x^2} = 2ax$$

$$\text{Differentiable at } x = 2 \Rightarrow \frac{-1}{4} = 4a \Rightarrow a = \frac{-1}{16}, b = \frac{3}{8}$$

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Solutions

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Q7

 f is continuous

$$f'(x) = 2x + 3, k < 1$$

$$\therefore 4 + a = b + 2$$

$$a = b - 2$$

$$b, x > 1$$

$$a = b - 2$$

 f is differentiable

$$\therefore b = 5$$

$$\therefore a = 3$$

$$\int_{-2}^1 (x^2 + 3x + 3) dx + \int_1^2 (5x + 2) dx$$

$$= \left[\frac{x^3}{3} + \frac{3x^2}{2} + 3x \right]_{-2}^1 + \left[\frac{5x^2}{2} + 2x \right]_1^2$$

$$= \left(\frac{1}{3} + \frac{3}{2} + 3 \right) - \left(\frac{-8}{3} + 6 - 6 \right) + \left(10 + 4 - \frac{5}{2} - 2 \right)$$

$$= 6 + \frac{3}{2} + 12 - \frac{5}{2} = 17$$

Q8

$$f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$$

$$f'(1) = 2024$$

$$f(1) = 1$$

Partially differentiating w. r. t. x

$$f'\left(\frac{x}{y}\right) \cdot \frac{1}{y} = \frac{1}{f(y)} f'(x)$$

$$y \rightarrow x$$

$$f'(1) \cdot \frac{1}{x} = \frac{f'(x)}{f(x)}$$

$$2024f(x) = xf'(x) \Rightarrow xf'(x) - 2024f(x) = 0$$

Q9

$$\text{Let } g(x) = ax + b$$

Now function $f(x)$ is continuous at $x = 0$

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Solutions

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$$\therefore \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}} = b$$

$$\Rightarrow 0 = b$$

$$\therefore g(x) = ax$$

Now, for $x > 0$

$$f'(x) = \frac{1}{x} \cdot \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}-1} \cdot \frac{1}{(2+x)^2}$$

$$+ \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}} \cdot \ln \left(\frac{1+x}{2+x} \right) \cdot \left(-\frac{1}{x^2} \right)$$

$$\therefore f'(1) = \frac{1}{9} - \frac{2}{3} \cdot \ln \left(\frac{2}{3} \right)$$

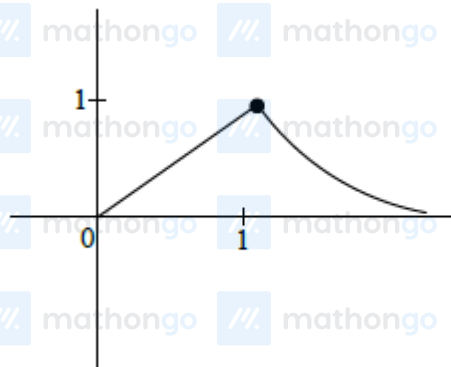
And $f(-1) = g(-1) = -a$

$$\therefore a = \frac{2}{3} \ln \left(\frac{2}{3} \right) - \frac{1}{9}$$

$$\therefore g(3) = 2 \ln \left(\frac{2}{3} \right) - \frac{1}{3}$$

$$= \ln \left(\frac{4}{9 \cdot e^{1/3}} \right)$$

Q10



$$f: (0, \infty) \rightarrow \mathbb{R}$$

$$f(x) = e^{-|\log_0 x|}$$

$$f(x) = \frac{1}{e^{|\ln x|}} = \begin{cases} \frac{1}{e^{-\ln x}}; & 0 < x < 1 \\ \frac{1}{e^{\ln x}}; & x \geq 1 \end{cases}$$

$$\begin{cases} \frac{1}{\frac{1}{x}} = x; & 0 < x < 1 \\ \frac{1}{x}, & x \geq 1 \end{cases}$$

 $m = 0$ (No point at which function is not continuous) $n = 1$ (Not differentiable)

$$\therefore m + n = 1$$

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Solutions

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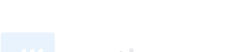
     

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